

2 Subject content and assessment information

Qualification aims and objectives

The sciences should be taught in ways that ensure that students have the knowledge to enable them to develop curiosity about the natural world, insight into working scientifically, and appreciation of the relevance of science to their everyday lives, so that they develop:

- scientific knowledge and conceptual understanding in the areas of biology, chemistry and physics
- understanding of the nature, processes and methods of science through scientific enquiry, which helps them to answer scientific questions on the world around them
- learning of how to apply observational, practical and enquiry-based skills to the world around them
- their ability to evaluate claims based on science through critical analysis of methods, evidence and conclusions, both qualitative and quantitative.

Scientific enquiry is set out separately but must always be taught through, and be clearly related to, the Pearson Edexcel International Award in Lower Secondary Science learning objectives.

Content

Overview

This Pearson Edexcel International Award in Lower Secondary Science requires students to demonstrate knowledge, understanding and application of the following learning objectives drawn from years 7, 8 and 9 of the Pearson Edexcel iLowerSecondary Curriculum in Science.

A full list of units that can be used in the assessment is given in *Appendix 3: Units*.

Content detail

Biology

Topic 1: Structure and function of living organisms – Life processes

Students should:		Guidance <i>iLowerSecondary Curriculum reference</i>
1.1	know the seven shared characteristics of living things	movement (even plants move, for example to track the Sun), respiration, sensitivity, growth, reproduction, excretion and nutrition <i>Year 7: Structure and function of living organisms – Life processes</i>
1.2	understand how the seven shared characteristics relate to a wide range of organisms in the local and the wider environment	<i>Year 7: Structure and function of living organisms – Life processes</i>
1.3	know the difference between breathing and respiration	<i>Year 8: Structure and function of living organisms – Life processes</i>
1.4	know that plant and animal cells respire to produce ATP to provide energy for cells	detailed knowledge of ATP is NOT required <i>Year 9: Structure and function of living organisms – Life processes</i>
1.5	know how to model aerobic and anaerobic respiration using word equations and a balanced symbol equation for aerobic respiration only	<i>Year 9: Structure and function of living organisms – Life processes</i>
1.6	know the structure of a virus and how it reproduces	<i>Year 9: Structure and function of living organisms – Life processes</i>
1.7	understand that viruses are obligate parasites causing harm to the cells of living things	<i>Year 9: Structure and function of living organisms – Life processes</i>
1.8	explain why viruses may not be classed as living organisms	<i>Year 9: Structure and function of living organisms – Life processes</i>